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Autonomous vehicle navigation of bypass merge lanes

Abstract

A system to improve the management of through traffic including an autonomous vehicle and other vehicles entering and exiting a multilane roadway. The lanes are separated by at least one lane separator which should not be crossed by traffic along a designated portion of the roadway, such as within a predetermined distance from a roadway entrance or exit. The at least one lane separator may be any lane marking, barrier, or the like. The at least one lane separator can be any length and located at any relative position with respect to the lanes.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 15/428,588 filed on Feb. 9, 2017 entitled BYPASS MERGE LANES, which claims the benefit of U.S. Provisional Patent Application No. 62/352,987 filed on Jun. 21, 2016, entitled BYPASS MERGE LANES, the disclosures of which are hereby incorporated herein in their entirety by this reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to the field of vehicular traffic management. Generally, the present invention relates to management of through traffic in the presence of vehicles entering and exiting a roadway to alleviate traffic congestion.

2. Description of the Prior Art

(2) Roadways such as freeways and highways typically have multiple lanes of traffic traveling in each direction. At certain locations along a multilane roadway, entrances and exits are provided to allow vehicles to enter and exit the roadway. Often other vehicles are traveling on the roadway, and vehicles entering the roadway must safely merge into traffic while vehicles exiting the roadway must safely maneuver to exit the roadway.

(3) Through traffic is often impeded due to traffic congestion in the vicinity of an entrance or exit of the roadway. Furthermore, vehicles entering the roadway into traffic can present a safety risk for both through traffic and the vehicles exiting the roadway. Also, vehicles exiting the roadway present a safety risk for both through traffic and vehicles entering the roadway. The risk is increased by motorists in vehicles changing lanes in the vicinity of an entrance or exit to avoid congested traffic.

(4) With the advent of technology to implement autonomous vehicles and the anticipated deployment of such vehicles, the problem of traffic being impeded due to traffic congestion in the vicinity of an entrance or exit of the roadway raises new concerns. The objective of the present invention addresses the navigation of autonomous vehicles through locations in the vicinity of an entrance or exit of the roadway having bypass merge lanes to safely alleviate traffic congestion.

SUMMARY OF THE INVENTION

(5) The present invention has as an object to provide a system to improve the management of through traffic in the vicinity of an entrance or exit where vehicles enter or exit a multilane roadway wherein one or more of the vehicles is an autonomous vehicle. Various examples of the system in accordance with the present invention manage vehicular traffic on a roadway with multiple lanes in at least one direction. The lanes are separated by one or more lane separators which should not be crossed by through traffic or vehicles entering or exiting the roadway along a designated portion of the roadway, such as within a predetermined distance from an entrance or exit provided for vehicles to enter or exit the roadway. The lane separators may be any lane marking, barrier, or the like. The lane separators may be any length and located at any designated relative positions with respect to the lanes. Additionally, signage is preferably provided to alert motorists of upcoming lane separators for managing traffic proximate an entrance or exit.

(6) In accordance with one aspect of the present invention, data is acquired for the positions of the lane separators. By way of example, the lane separator data may be acquired by an autonomous vehicle, such as, by means of a camera mounted on the autonomous vehicle, when the vehicle encounters a location of the roadway having lane separators. In accordance with another example of the present invention, the data for the positions of the lane separators may be acquired by means of Global Positioning System (GPS) satellite images. In accordance with yet another example of the present invention, the data for the positions of the lane separators may be provided by the

transportation authority which implements the lane separators.

(7) In accordance with the examples, the data for the positions of the lane separators are preferably stored in a database. An autonomous vehicle is provided with access to the database to download the data. The downloaded data is utilized by the autonomous vehicle to navigate the lane separators based on the downloaded data.

(8) In one preferred example, after the positions of the lane separators have been acquired, the data for the positions of the lane separators are stored in a lookup table in a database. A transmitter is mounted on the signage to transmit an identification of the upcoming location of the lane separators as the autonomous vehicle approaches the location. The autonomous vehicle is equipped with a transponder to receive the transmitted identification and transmits the identification to the lookup table which is used to look up the locations of the lane separators and download the data for the lane separators to the autonomous vehicle. The autonomous vehicle then utilizes the data for the lane separators to navigate the upcoming lane separators.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) The various examples of the present invention will be described in conjunction with the accompanying figures of the drawing to facilitate an understanding of the present invention. In the drawing:

(2) FIG. 1 is an elevational view of a multilane roadway for management of vehicular traffic at one or more entrances and exits to a roadway in accordance with an example of the present invention;

(3) FIG. 2 is an elevational view of another example of a multilane roadway for management of vehicular traffic at one or more entrances and exits to a roadway in accordance with the present invention;

(4) FIG. 3 is an elevational view of an additional example of a multilane roadway for management of vehicular traffic at one or more entrances and exits to a roadway in accordance with the present invention;

(5) FIG. 4 is an elevational view of a further example of a multilane roadway for management of vehicular traffic at one or more entrances and exits to a roadway in accordance with the present invention;

(6) FIG. 5 is an elevational view of an alternative example of the multilane roadway for management of vehicular traffic at one or more entrances and exits to the roadway shown in FIG. 4;

(7) FIG. 6 is an elevational view of an example of a multilane roadway for management of vehicular traffic at one or more entrances and exits to a roadway proximate roadway junctions in accordance with the present invention;

(8) FIG. 7 is an elevational view of another example of a multilane roadway for management of vehicular traffic at one or more entrances and exits to a roadway proximate roadway junctions in accordance with the present invention;

(9) FIG. 8 is an elevational view of an additional example of a multilane roadway for management of vehicular traffic at one or more entrances and exits to a roadway proximate roadway junctions in accordance with the present invention;

(10) FIG. 9, comprising FIGS. 9A to 9C, illustrates examples of signage to alert motorists to upcoming designated traffic management portions of a roadway in accordance with the present invention;

(11) FIG. 10 illustrates an example of a system to acquire and store data for the positions of lane separators in accordance with the present invention; and

(12) FIG. 11 illustrates a flow chart for navigating the positions of lane separators by an

autonomous vehicle based on the data acquired for the positions of lane separators.

DETAILED DESCRIPTION OF THE PREFERRED EXAMPLES

(13) An example of a multilane roadway management system **10** in accordance with the present invention is shown in FIG. **1**. A roadway **11** comprises a plurality of lanes **12** in one direction, for example, six lanes, as shown in FIG. **1**. The lanes **12** are preferably marked by lane marking **14** to define each of the lanes, for example, by conventional striping in the form of dashed line segments to designate the lanes.

(14) As shown in FIG. **1**, one or more entrances **16** are provided to allow vehicles to enter the roadway **11**. In accordance with the example of the present invention shown in FIG. **1**, one or more lane separators **18** is provided to divide the roadway **11** into one or more merge lanes **20** and one or more bypass lanes **22** proximate to the one or more entrances **16** or exits **24**. The one or more lane separators **18** should not be crossed by through traffic or merging or exiting vehicles along the portion of the roadway where the lane separator(s) is/are present. The one or more lane separators **18** may be any lane marking, barrier, or the like. The one or more lane separators **18** may be any length and located at any position relative to the lanes **12**. As shown in FIG. **1**, for example, the one or more lane separators **18** may be a lane marking consisting of a solid line of striping distinguishable from the dashed line segments **14** of striping that define the lanes **12**.

(15) The multilane roadway management system **10** provides one or more lane separators **18** which vehicles should not cross between the bypass lanes **22** designated for through traffic and the merge lanes **20** to reduce traffic congestion and to improve the safety of the roadway **11** for both through traffic and vehicles entering and exiting the roadway at the entrances **16** and exits **24**. The merge lanes **20** are those lanes proximal to an entrance **16** (i.e., closer to an entrance) or exit **24** (i.e., closer to an exit) for vehicles entering or exiting the roadway **11**, respectively. The bypass lanes **22** are those lanes distal from the entrance **16** (i.e., farther from the entrance) or exit **24** (i.e., farther from the exit) on the other side of the merge lanes **20** for through traffic on the roadway **11**.

(16) As shown in FIG. **1**, in accordance with one example of the present invention, one or more lane separators **18A** may be provided to separate the lanes **12** into merge lanes **20A** and bypass lanes **22A** at one or more predetermined locations along the roadway **11**. In accordance with another example, lane separators **18B** may partially overlap to alter the number of merge lanes **20B** and bypass lanes **22B** at one or more other predetermined locations along the roadway **11**. In accordance with a further example, lane separators **18C** may entirely overlap at one or more predetermined locations along the roadway **11** to manage through traffic into merge lanes **20C** and bypass lanes **22C**, with the lane separators **18C** establishing multiple corridors (e.g., two corridors) of bypass lanes **22C1** and **22C2**.

(17) If there is an exit **24** in proximity to one or more of the entrances **16**, the multilane roadway management system **10** preferably provides appropriate roadway signage to alert motorists approaching the exit that they need to maneuver their vehicles into the merge lanes **20** and be prepared to exercise caution, including reducing their speed, to safely allow vehicles entering the roadway to merge into traffic at the entrance(s) **16** while allowing vehicles exiting the roadway to safely exit without impeding through traffic.

(18) Referring to FIG. **2**, an example of the multilane roadway management system **10** is shown where multiple entrances **16** and exits **24** are spaced along the roadway **11**. As shown in FIG. **2**, lane separators **18** having various lengths divide the roadway **11** into merge lanes **20** and bypass lanes **22** in the vicinity of the entrances **16** and exits **24**. The bypass lanes **22** begin where the conventional striping in the form of dashed line segments **14** to define the lanes **12** of the multilane roadway **11** transitions to lane marking consisting of a solid line, for example. The positioning of the lane separators **18** is determined by the spacing of the entrances **16** to allow vehicles entering the roadway **11** to safely merge into through traffic while accommodating vehicles entering the roadway at a subsequent entrance **16** to merge onto the roadway after initially entering an additional entrance/exit lane **26**. In instances where an entrance such as entrance **16A** is in close

proximity to an exit such as exit **24A**, the number of merge lanes **20** is sufficient to allow entering and exiting vehicles to safely enter and exit the roadway **11** with minimal effect on through traffic which is restricted to bypass lanes **22**.

(19) Referring to FIG. **3**, an example of the multilane roadway management system **10** is shown where multiple entrances **16** and exits **24** are spaced along the roadway **11**. As shown in FIG. **3**, lane separators **18** having various lengths divide the roadway **11** into merge lanes **20** and bypass lanes **22** in the vicinity of the entrances **16** and exits **24**. The bypass lanes **22** begin where the conventional striping in the form of dashed line segments **14** to define the lanes **12** of the multilane roadway **11** transitions to lane marking consisting of a solid line, for example. The positioning of the lane separators **18** is determined by the spacing of the entrances **16** to allow vehicles entering the roadway **11** to safely merge into through traffic while accommodating vehicles entering the roadway at a subsequent entrance **16** to safely merge onto the roadway after initially entering an additional entrance/exit lane **28**. In instances where an entrance such as entrance **16A** is in close proximity to an exit such as exit **24A**, the number of merge lanes **20** is sufficient to allow entering and exiting vehicles to enter and exit the roadway **11** with minimal effect on through traffic which is restricted to bypass lanes **22**.

(20) Referring to FIG. **4**, an example of the multilane roadway management system **10** is shown where multiple entrances **16** and exits **24** are spaced along the roadway **11** and a diamond lane **30** is also present to allow vehicles to exit the roadway at a flyover **31** or continue on in the diamond lane. As shown in FIG. **4**, lane separators **18** having various lengths divide the roadway **11** into merge lanes **20** and bypass lanes **22** including the diamond lane **30** in the vicinity of the entrances **16** and exits **24**. The bypass lanes **22** begin where the conventional striping in the form of dashed line segments **14** to define the lanes **12** of the multilane roadway **11** transitions to lane marking consisting of a solid line, for example. The positioning of the lane separators **18** is determined by the spacing of the entrances **16** to allow vehicles entering the roadway **11** to safely merge into through traffic while accommodating vehicles entering the roadway at a subsequent entrance **16** to safely merge onto the roadway after initially entering additional entrance/exit lanes **28**. In instances where the entrances **16** are in close proximity to the exits **24**, the number of merge lanes **20** is sufficient to allow entering and exiting vehicles to safely enter and exit the roadway **11** after first entering the additional entrance/exit lanes **28** with minimal effect on through traffic which is restricted to bypass lanes **22**.

(21) Referring to FIG. **5**, an example of the multilane roadway management system **10** is shown where multiple entrances **16** and exits **24** are spaced along the roadway **11** and a diamond lane **30** is also present to allow vehicles to exit the roadway. Furthermore, a diamond lane **32** is established by a roadway marking **34** for through traffic on the roadway **11**. As shown in FIG. **5**, lane separators **18** having various lengths divide the roadway **11** into merge lanes **20** and bypass lanes **22** including the diamond lanes **30** and **32** in the vicinity of the entrances **16** and exits **24**. The bypass lanes **22** begin where the conventional striping in the form of dashed line segments **14** to define the lanes **12** of the multilane roadway **11** transitions to lane marking consisting of a solid line, for example. The positioning of the lane separators **18** is determined by the spacing of the entrances **16** to allow vehicles entering the roadway **11** to safely merge into through traffic while accommodating vehicles entering the roadway at a subsequent entrance **16** to merge onto the roadway after initially entering additional entrance/exit lanes **28**. In instances where the entrances **16** are in close proximity to the exits **24**, the number of merge lanes **20** is sufficient to allow entering and exiting vehicles to safely enter and exit the roadway **11** after first entering the additional entrance/exit lanes **28** with minimal effect on through traffic which is restricted to bypass lanes **22**.

(22) Referring to FIG. **6**, an example of the multilane roadway management system **10** is shown where the roadway **11** comprises a first multilane roadway **40** which merges with a second multilane roadway **42** and multiple entrances **16** and exits **24** are spaced along the roadway **42**. As shown in FIG. **6**, lane separators **18** having various lengths divide the roadways **40** and **42** into

merge lanes **20** and bypass lanes **22** where the roadways **40** and **42** merge and in the vicinity of the entrances **16** and exits **24**. The bypass lanes **22** begin where the conventional striping in the form of dashed line segments **14** to define the lanes **12** of the multilane roadways **40** and **42** transitions to lane marking consisting of a solid line, for example. The positioning of the lane separators **18** is determined by the merging of the multilane roadways **40** and **42** and spacing of the entrances **16** along the multilane roadway **42** to allow vehicles to merge from one of the multilane roadways **40** and **42** onto the other of the multilane roadways, while accommodating vehicles entering the multilane roadway **42** at entrances **16** to merge onto the roadways **40** and **42** after first entering additional entrance/exit lanes **28**. In instances where the entrances **16** are in close proximity to the exits **24**, the number of merge lanes **20** is sufficient to allow entering and exiting vehicles to safely enter and exit the multilane roadway **42** after entering the additional entrance/exit lanes **28** with minimal effect on through traffic which is restricted to bypass lanes **22**.

(23) Referring to FIG. 7, an example of the multilane roadway management system **10** is shown where the roadway **11** comprises a first multilane roadway **44** which splits into a second multilane roadway **46** and a third multilane roadway **48**. Multiple entrances **16** and exits **24** are spaced along the multilane roadway **44**, and a diamond lane **30** is also present. As shown in FIG. 7, lane separators **18** having various lengths divide the multilane roadway **44** into merge lanes **20** and bypass lanes **22** including the diamond lane **30** in the vicinity of the entrances **16** and exits **24**. The bypass lanes **22** begin where the conventional striping in the form of dashed line segments **14** to define the lanes **12** of the multilane roadway **44** transitions to lane marking consisting of a solid line, for example. The positioning of the lane separators **18** is determined by the spacing of the entrances **16** to allow vehicles entering the multilane roadway **44** to safely merge into through traffic while accommodating vehicles entering the multilane roadway **44** at a subsequent entrance **16** to merge onto the multilane roadway after first entering an additional entrance/exit lane **28**. In instances where the entrances **16** are in close proximity to exits **24**, the number of merge lanes **20** is sufficient to allow entering and exiting vehicles to safely enter and exit the multilane roadway **44** after first entering the additional entrance/exit lane **28** with minimal effect on through traffic which is restricted to bypass lanes **22**.

(24) Referring to FIG. 8, an example of the multilane roadway management system **10** is shown where the roadway **11** comprises a first multilane roadway **50** which splits into a second multilane roadway **52**, a third multilane roadway **54**, and a fourth multilane roadway **56**. Multiple entrances **16** and exits **24** are spaced along the multilane roadway **50**, and a diamond lane **30** is also present on roadway **50**. As shown in FIG. 8, lane separators **18** having various lengths divide the multilane roadway **50** into merge lanes **20** and bypass lanes **22** including the diamond lane **30** in the vicinity of the entrances **16** and exits **24**. The bypass lanes **22** begin where the conventional striping in the form of dashed line segments **14** to define the lanes **12** of the multilane roadway **50** transitions to lane marking consisting of a solid line, for example. The positioning of the lane separators **18** is determined by the spacing of the entrances **16** to allow vehicles entering the multilane roadway **50** to safely merge into through traffic while accommodating vehicles entering the multilane roadway **50** at a subsequent entrance **16** to merge onto the multilane roadway **50** after first entering an additional entrance/exit lane **28**. In instances where the entrances **16** are in close proximity to the exits **24**, the number of merge lanes **20** is sufficient to allow entering and exiting vehicles to safely enter and exit the multilane roadway **50** after entering the additional entrance/exit lane **28** with minimal effect on through traffic which is restricted to bypass lanes **22**. As shown in FIG. 8, note that the lane separator **18A** may promote safety by restricting vehicles from entering the multilane roadway **52** after initially merging onto the multilane roadway **50**.

(25) Referring to FIGS. 9A, 9B, and 9C, signage is preferably provided to alert motorists of approaching exits and related restrictions on lane changes imposed by lane separators **18** to reduce congestion by funneling through traffic into bypass lanes **22** and apprising motorists who want to exit to maneuver into merge lanes **20**. As shown in FIG. 9A, signage may comprise a first sign **60**

to alert motorists that they are approaching a bypass lane(s) for through traffic and the distance to the bypass lane(s). A second sign **62** may be existing signage informing motorists about exits which are upcoming and the distances to those exits, as well as the distance to an upcoming roadway junction.

(26) As shown in FIG. **9B**, signage may comprise a plurality of elements that are either separate signs or integrated into one sign. The signage may comprise a first element **64** to alert motorists that they are approaching a bypass lane(s) for through traffic and the distance to the bypass lane(s). A second element **66** may be signage informing motorists about local exits which are upcoming and the distances to those exits. A third element **68** may be signage informing motorists about a roadway junction which is upcoming and the distance to the roadway junction. Note that the distance to the bypass lane(s) is less than the distance to the local exits or the roadway junction, and therefore the signage respecting the bypass lane(s) appears at the top.

(27) As shown in FIG. **9C**, signage may comprise a plurality of elements that are either separate signs or integrated into one sign. The signage may comprise a first element **70** which may be signage informing motorists about local exits which are upcoming and the distances to those exits. A second element **72** may be signage informing motorists about a roadway junction which is upcoming and the distance to the roadway junction. A third element **72** may alert motorists that they are approaching a bypass lane(s) for through traffic and the distance to the bypass lane(s). Note that the distance to the bypass lane(s) appears at the bottom to give priority to the signage for the local exits and roadway junction.

(28) The examples described above are provided by way of example only, and various modifications apparent to persons skilled in the art are contemplated. For example, as shown in FIG. **8**, instead of striping, the lane separator **18A** may be a barrier installed between the lanes **12** of the roadway **11** which are vertically movable from a position with the top of the barrier flush with the surface of the roadway to a raised position extending above the surface of the roadway. Alternatively, as shown in FIG. **1**, the lane separator **18A** may be a moveable road zipper barrier commercially available from Lindsay Corporation located in Rio Vista, California.

(29) Referring again to FIGS. **1-8**, in accordance with one aspect of the present invention, data is acquired for the positions of the lane separators **18**. By way of example, as shown in FIG. **10**, the positions of the lane separators **18** may be acquired by an autonomous vehicle **80**, such as by means of a camera(s) **82** mounted on the autonomous vehicle to obtain lane separator data, when the vehicle encounters a portion of the roadway **11** having lane separators. In accordance with another example of the present invention, data for the positions of the lane separators **18** may be acquired from images obtained by a Global Positioning System (GPS) satellite camera(s) **90**. In accordance with yet another example of the present invention, data for the positions of the lane separators **18** may be provided by a transportation authority **100** which implements the lane separators.

(30) In accordance with the examples, the data for the positions of the lane separators **18** are preferably stored in a database **110**. The autonomous vehicle **80** is provided with access to the database **110** by means of a wireless modem **120** to download the data for the positions of the lane separators **18**. The downloaded lane separator data is utilized by an autonomous in-vehicle computer **122** of the autonomous vehicle **80** to navigate the lane separators **18** based on the downloaded lane separator data as will be described below.

(31) In one preferred example, after the positions of the lane separators **18** have been acquired, the data for the positions of the lane separators are stored in a lookup table **110A** comprising the database **110**. As shown in FIG. **9A**, a transmitter **61** is preferably mounted on the signage **60** to transmit an identification code of the upcoming positions of the lane separators **18** as the autonomous vehicle **80** approaches the location where the lane separators are present. The autonomous vehicle **80** is equipped with a transponder **130** to receive the transmitted identification code and transmits the identification code to the lookup table **110A** which is used to look up the

positions of the lane separators **18** and download the data for the lane separators to the autonomous vehicle via the wireless modem **120**. The autonomous vehicle **80** then utilizes the data for the lane separators **18** to navigate the upcoming lane separators. In accordance with a modification of the example of the present invention, the transponder **130** may be incorporated into the wireless modem **120**.

(32) FIG. **11** illustrates a flow chart for the autonomous vehicle **80** navigating the positions of lane separators **18** based on the data acquired for the positions of lane separators. The autonomous in-vehicle computer **122** shown in FIG. **10** comprises a processor which is connected to program and data memory **124** to access instructions and data for navigating the positions of lane separators **18**. For example, the autonomous in-vehicle computer **122** may be an ACO-3000-BRD-U in-vehicle computer commercially available from Premio located in City of Industry, California or an ABOX-5200 G-4 in-vehicle computer commercially available from Sintrones located in New Taipei City, Taiwan.

(33) As shown in FIG. **11**, the autonomous in-vehicle computer **122** shown in FIG. **10** is activated at step **200** when the autonomous vehicle **80** is placed in operation. Preferably, the instructions executed by the autonomous in-vehicle computer **122** for navigating the positions of lane separators **18** are instantiated when a destination for the autonomous vehicle **80** is entered, as indicated at step **210** shown in FIG. **11**.

(34) As indicated at step **220** in FIG. **11**, the instructions executed by the autonomous in-vehicle computer **122** may selectively enable the instructions to continue, as indicated at step **220** shown in FIG. **11**. For example, the transportation authority **100** may require that the lane separators **18** are in force only at certain times, for example, during rush hours. If not required, the instructions executed by the autonomous in-vehicle computer **122** end, as indicated at step **230**.

(35) If lane separators **18** are currently in force prior to reaching the entered destination, the instructions executed by the autonomous in-vehicle computer **122** continue and receive continuous input of the present location of the autonomous vehicle **80**, as indicated at step **240** shown in FIG. **11**. For example, the present location of the autonomous vehicle **80** may be obtained using conventional GPS data received by the autonomous in-vehicle computer **122** via the wireless modem **120**.

(36) As indicated at step **250** shown in FIG. **11**, the instructions executed by the autonomous in-vehicle computer **122** preferably determine whether the destination entered at step **210** is beyond any intervening location where lane separators **18** are currently in force. If so, the instructions executed by the autonomous in-vehicle computer **122** control the autonomous vehicle **80** to be in one of the through lanes (i.e., one of the bypass lanes **22**) to the entered destination, as indicated at step **260** and returns to step **220**.

(37) If the entered destination is less than or equal to a given distance x beyond the lane separators **18** currently in force as indicated at step **270** shown in FIG. **11**, the instructions executed by the autonomous in-vehicle computer **122** control the position(s) the autonomous vehicle **80** to safely navigate into one of the right hand lanes to proceed to exit, as indicated at step **280** in FIG. **11**. The autonomous vehicle **80** then exits and proceeds to the entered destination and the instructions executed by the autonomous in-vehicle computer **122** end, as indicated at steps **290** and **300** shown in FIG. **11**, respectively.

(38) If, on the other hand, the entered destination is greater than the given distance x as indicated at step **270**, the instructions executed by the autonomous in-vehicle computer **122** control the position(s) the autonomous vehicle **80** to navigate the vehicle to safely proceed toward the entered destination, as indicated at step **310** and then returns to step **270** as shown in FIG. **11**.

(39) The example described in FIG. **11** is provided by way of example only, and various modifications apparent to persons skilled in the art are contemplated.

Claims

1. A system to manage vehicular traffic on a roadway having multiple through lanes of vehicular traffic moving in the same direction to reduce congestion in the vicinity of an entrance enabling vehicles to enter the roadway, the roadway not having a High Occupancy Vehicle (HOV) or Express Lane in the vicinity of the entrance, comprising in combination: a roadway comprising a plurality of m through lanes of vehicular traffic moving in the same direction in the vicinity of an entrance to the roadway where m is an integer and $m > 2$, wherein the m through lanes are delineated by dashed line segments having a given length between the lanes; and a system comprising a plurality of lane separators positioned between n of the m through lanes where n is an integer and $2 \leq n \leq m$, which should not be crossed by vehicular traffic along a designated portion of the roadway within a predetermined distance d from the entrance to the roadway; wherein the plurality of lane separators differ from dashed line segments delineating the m through lanes, the lane separators having a plurality of diverse predetermined lengths different from the given length and being located at predetermined positions with respect to the m through lanes to indicate that vehicular traffic should not cross any lane separator from one lane of the m through lanes in which the vehicular traffic is within the predetermined distance d from the entrance to an adjacent lane where one of the lane separators is positioned between the one lane and the adjacent lane; wherein the plurality of lane separators define at least one of the m through lanes as an entrance bypass lane distal to the entrance to the roadway and at least one of the m through lanes as a merge lane proximal to the entrance to the roadway to enable vehicles to merge at the entrance into the at least one merge lane; and wherein the plurality of lane separators are located amongst the m through lanes to manage entry of vehicles into the at least one merge lane at the entrance to the roadway; the combination further comprising: an autonomous vehicle to navigate the roadway, the autonomous vehicle comprising: an in-vehicle computer to execute instructions responsive to lane separator data to control the position of the autonomous vehicle to navigate the lane separators such that the autonomous vehicle does not cross any lane separator from one lane of the m through lanes in which the autonomous vehicle is within the predetermined distance d from the entrance to an adjacent lane where one of the lane separators is positioned between the one lane and the adjacent lane.
2. The system as recited in claim 1 wherein the plurality of lane separators are in the form of solid lines of striping.
3. The system as recited in claim 1 wherein the plurality of lane separators comprise at least one barrier that is vertically movable from a position with a top of the at least one barrier flush with a surface of the roadway to a raised position extending above the surface of the roadway.
4. The system as recited in claim 1 wherein the plurality of lane separators comprise at least one moveable road zipper barrier.
5. The system as recited in claim 1, further comprising an additional entrance lane adjacent to the at least one merge lane.
6. The system of claim 1 wherein the autonomous vehicle further comprises at least one camera to acquire the lane separator data.
7. The system of claim 1 wherein the lane separator data is obtained based on a satellite image.
8. The system of claim 1 wherein the lane separation data is provided by a transportation authority.
9. The system as recited in claim 1, further comprising signage to alert motorists that they are approaching the at least one entrance bypass lane for through traffic and a distance to the at least one entrance bypass lane.
10. The system as recited in claim 9, further comprising a transmitter mounted on the signage to transmit an identification code to the autonomous vehicle, and the autonomous vehicle further comprises a transponder to receive the identification code and transmit the identification code to a

lookup table, and the lookup table transmits the lane separator data to the autonomous vehicle.

11. A system to manage vehicular traffic on a roadway having multiple through lanes of vehicular traffic moving in the same direction to reduce congestion in the vicinity of an entrance enabling vehicles to enter the roadway, the roadway having at least one High Occupancy Vehicle (HOV) or Express Lane in the vicinity of the entrance and additionally having a plurality of in through lanes of vehicular traffic moving in the same direction as the traffic moving in the at least one HOV or Express Lane in the vicinity of the entrance to the roadway where m is an integer and $m > 2$, wherein the m through lanes are delineated by dashed line segments having a given length between the lanes, the improvement comprising in combination: a system comprising a plurality of lane separators positioned between n of the m through lanes where n is an integer and $2 \leq n \leq m$, which should not be crossed by vehicular traffic along a designated portion of the roadway within a predetermined distance d from the entrance to the roadway; wherein the plurality of lane separators differ from dashed line segments delineating the m through lanes, the lane separators having a plurality of diverse predetermined lengths different from the given length and being located at predetermined positions with respect to the m through lanes to indicate that vehicular traffic should not cross any lane separator from one lane of the m through lanes in which the vehicular traffic is within the predetermined distance d from the entrance to an adjacent lane where one of the lane separators is positioned between the one lane and the adjacent lane; wherein the plurality of lane separators define at least one of the m through lanes as an entrance bypass lane distal to the entrance to the roadway and at least one of the m through lanes as a merge lane proximal to the entrance to the roadway to enable vehicles to merge at the entrance into the at least one merge lane; and wherein the plurality of lane separators are located amongst the m through lanes to manage entry of vehicles into the at least one merge lane at the entrance to the roadway; and the combination further comprising: an autonomous vehicle to navigate the roadway, the autonomous vehicle comprising: an in-vehicle computer to execute instructions responsive to lane separator data to control the position of the autonomous vehicle to navigate the lane separators such that the autonomous vehicle does not cross any lane separator from one lane of the m through lanes in which the autonomous vehicle is within the predetermined distance d from the entrance to an adjacent lane where one of the lane separators is positioned between the one lane and the adjacent lane.

12. The system as recited in claim 11 wherein the plurality of lane separators are in the form of solid lines of striping.

13. The system as recited in claim 11 wherein the plurality of lane separators comprise at least one barrier that is vertically movable from a position with a top of the at least one barrier flush with a surface of the roadway to a raised position extending above the surface of the roadway.

14. The system as recited in claim 11 wherein the plurality of lane separators comprise at least one moveable road zipper barrier.

15. The system as recited in claim 11, further comprising an additional entrance lane adjacent to the at least one merge lane.

16. The system of claim 11 wherein the autonomous vehicle further comprises at least one camera to acquire the lane separator data.

17. The system of claim 11 wherein the lane separator data is obtained based on a satellite image.

18. The system of claim 11 wherein the lane separation data is provided by a transportation authority.

19. The system as recited in claim 11, further comprising signage to alert motorists that they are approaching the at least one entrance bypass lane for through traffic and a distance to the at least one entrance bypass lane.

20. The system as recited in claim 19, further comprising a transmitter mounted on the signage to transmit an identification code to the autonomous vehicle, and the autonomous vehicle further

comprises a transponder to receive the identification code and transmit the identification code to a lookup table, and the lookup table transmits the lane separator data to the autonomous vehicle.
